

Topics

Data splitting (train, validation, test)

Cross validation and types

Overfitting

Solutions for overfitting problem

Underfitting

Regularization (Dropout , Augmentation)

Data splitting



Training Dataset

Training Dataset: The sample of data used to fit the model.

The actual dataset that we use to train the model (weights and biases in the case of a Neural Network).

The model sees and learns from this data.

Validation Dataset or development set

- ❑ **The validation set** is used to evaluate a given model, but this is for frequent evaluation.
- ❑ We, as machine learning engineers, use this data to **fine-tune the model** hyperparameters.
- ❑ We use the **validation set results**, and update higher level **hyperparameters**.
- ❑ So the validation set affects a model, but only indirectly.

Test Dataset

Test Dataset: The sample of data used to provide an evaluation of a final model fit on the training dataset.

The Test dataset provides the gold standard used to evaluate the model. It is only used once a model is completely trained(using the train and validation sets).

Cross-Validation

Cross-Validation also referred to as **out of sampling technique** is an essential element of a data science project.

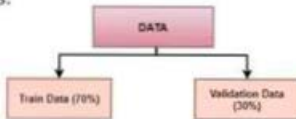
It is a resampling procedure used to evaluate machine learning models and access how the model will perform for an independent test dataset.

Three types of cross validation:

1. Holdout cross-validation.
2. k-fold cross-validation.
3. Repeated random subsampling validation.

Holdout cross-validation:

The holdout technique is an exhaustive cross-validation method, that randomly splits the dataset into train and test data depending on data analysis.



In the case of holdout cross-validation, the dataset is randomly split into training and validation data. Generally, the split of training data is more than test data. The training data is used to induce the model and validation data is evaluates the performance of the model.

The more data is used to train the model, the better the model is. For the holdout cross-validation method, a good amount of data is isolated from training.

Holdout cross-validation:

Pros:

1. Simple, easy to understand, and implement.

Cons:

1. Not suitable for an **imbalanced** dataset.
2. A lot of data is isolated from training the model.

Hint: a dataset is said to be imbalanced when there is a significant, or in some cases extreme, disproportion among the number of examples of each class of the problem.

k-fold cross-validation:

In k-fold cross-validation, the original dataset is equally partitioned into k subparts or folds. Out of the k-folds or groups, for each iteration, one group is selected as validation data, and the remaining (k-1) groups are selected as training data.

$n = 12$
 $k = 3$

Data



Test



Train



The process is repeated for k times until each group is treated as validation and remaining as training data.

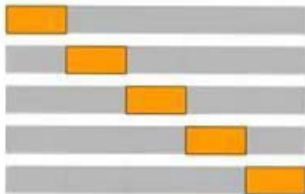
k-fold cross-validation:

In k-fold cross-validation, the original dataset is equally partitioned into k subparts or folds. Out of the k-folds or groups, for each iteration, one group is selected as validation data, and the remaining (k-1) groups are selected as training data.

$n = 12$
 $k = 3$
Shuffled
Data

Test

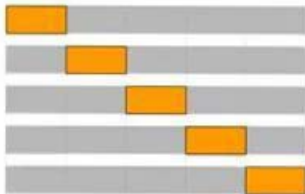
Train



The process is repeated for k times until each group is treated as validation and remaining as training data.

k-fold cross-validation:

In k-fold cross-validation, the original dataset is equally partitioned into k subparts or folds. Out of the k-folds or groups, for each iteration, one group is selected as validation data, and the remaining (k-1) groups are selected as training data.



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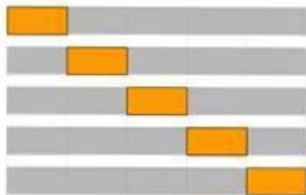
$n = 12$

$k = 2$

Model 2

Test

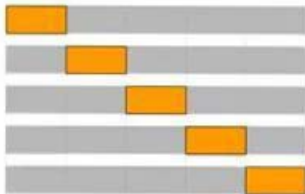
Train



The process is repeated for k times until each group is treated as validation and remaining as training data.

k-fold cross-validation:

In k-fold cross-validation, the original dataset is equally partitioned into k subparts or folds. Out of the k-folds or groups, for each iteration, one group is selected as validation data, and the remaining (k-1) groups are selected as training data.



The process is repeated for k times until each group is treated as validation and remaining as training data.

k-fold cross-validation:

The final accuracy of the model is computed by taking the mean accuracy of the k-models validation data.

$$acc_{cv} = \sum_{i=1}^k \frac{acc_i}{k}$$

Pros:

1. Low time complexity
2. The entire dataset is utilized for both training and validation.

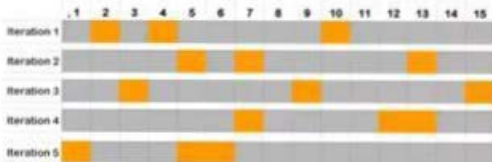
Cons:

1. Not suitable for an imbalanced dataset.

Repeated random subsampling validation:

- ❑ Repeated random subsampling validation also referred to as Monte Carlo cross-validation splits the dataset randomly into training and validation.
- ❑ Unlike k-fold cross-validation split of the dataset into not in groups or folds but splits in this case in random.
- ❑ The number of iterations is not fixed and decided by analysis.
- ❑ The results are then averaged over the splits.

Repeated random subsampling validation:



Pros:

1. The proportion of train and validation splits is not dependent on the number of iterations or partitions.

Cons:

1. Some samples may not be selected for either training or validation.

2. Not suitable for an imbalanced dataset.